Dialectica Constructions over Monoidal Bases

Valeria de Paiva
Topos Institute, Berkeley

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Abstract

The Dialectica construction, the categorical counterpart of Gödel's Dialectica interpretation, gives models of linear logic that have been used in proof theory, set theory, game semantics, concurrency and dependent type theory. It was first given over a cartesian closed category, with relations as subobjects. Over sets, relations can be given as subobjects or as maps into the booleans. This note takes relations valued in an object of truth values, a lineale, and also generalizes the construction a little. The base needs only to be a symmetric monoidal closed category instead of a cartesian closed one, as for the Chu construction. Graded sets give an example that is not cartesian and carries the full structure of a model of intuitionistic linear logic.

1 Introduction

The Dialectica construction is the categorical counterpart of Gödel's Dialectica interpretation [5]: a formula is read as a relation between witnesses and counterexamples, and a proof of an implication as a pair of functionals, one carrying witnesses forward and one carrying counterexamples back. The thesis [2] gives two such constructions over a cartesian closed category $\mathcal{C}$. In Dial, as in Gödel's interpretation, the backward map reads the witness, and $\mathsf{Dial}(\mathcal{C})$ models intuitionistic linear logic; in the Girard variant Gir it does not, and $\mathsf{Gir}(\mathcal{C})$ models classical linear logic. This note is about Dial. Its objects are subobjects $\alpha \rightarrowtail U \times X$, with U an object of witnesses and X an object of counterexamples, and a morphism is a map $f\colon U \to V$ forward on witnesses together with a map $F\colon U \times Y \to X$ backward on counterexamples, such that α pulled back along $\langle \pi_1, F\rangle$ is contained in β pulled back along $f \times \mathrm{id}_Y$. In Gir the backward map is $F\colon Y \to X$ instead.

The construction has been taken up in several directions. Hyland [17] placed it in an abstract account of proof theory; Bodil Biering [18] and Hofstra [15] recast it fibrationally; and Trotta, Spadetto and I studied the logical principles it validates, through Gödel doctrines [19, 16]. Tamara von Glehn and Sean Moss [20] extended the Dialectica pattern of witnesses and counterexamples to models of dependent type theory, in a version without predicates. Blass [21] found a variant of the Girard construction over **Set** in set theory, where its morphisms compare cardinal characteristics of the continuum, and Hedges [22] read its morphisms as strategies in games with bidding. Outside logic it has served as a template for models of concurrency, from Brown and Gurr's Petri nets [14] to the recent work of Elena Di Lavore, Wilmer Leal and myself [23], where the truth values are a lineale such as the natural numbers rather than the booleans.

Our reason for leaving the cartesian closed base was the category **FinVect** of finite-dimensional vector spaces over a field K. Lafont and Streicher [9] observed that these spaces live inside Chu's construction over K, and we wanted the corresponding picture for Dial. Since **FinVect** is symmetric monoidal closed with products but not cartesian closed, this requires the construction over such a base. There are two steps away from the thesis. A companion note [7] replaces the subobjects by maps $\alpha\colon U \times X \to \Omega$ into a lineale [8] internal to $\mathcal{C}$, still over a cartesian closed base. The present note takes the second step, replacing $\mathcal{C}$ by a symmetric monoidal closed category $\mathcal{V}$ with finite products, so that relations become maps $U \otimes X \to \Omega$. For Gir this was

done in [3], and the article [1] returned to it; for Dial the calculations were never written down. Besides the internal lineale Ω (Definition 3.1), the base must supply witnesses that are comonoids and forward maps that are comonoid morphisms. A cartesian base hides both requirements, since there every object is a comonoid with its diagonal and every map preserves this structure; they are needed because the backward map of a dialectica morphism uses the witness twice, once to produce a counterexample and once to evaluate the relation, so composition has to copy witnesses (Section 3).

What is new here is a concrete example. The category **GrSet** of $\mathbb{N}$-graded sets, families $X = (X_n)_{n\in\mathbb{N}}$ of sets, with the Cauchy product $(X \otimes Y)_n = \coprod_{i+j=n} X_i \times Y_j$, is symmetric monoidal closed but not cartesian, since its unit, a one-point set in degree 0, is not terminal (Section 4). Its comonoids are the graded sets concentrated in degree 0, that is, plain sets, every map between them is a comonoid morphism, and any lineale placed in every degree is an internal lineale. An object of $\mathsf{Dial}_\Omega(\mathbf{GrSet})$ is then a set of witnesses U, a graded set of counterexamples X and a relation $\alpha_n \colon U \times X_n \to \Omega$ in each degree, and a morphism is a function $f \colon U \to V$ with backward maps $F_n \colon U \times Y_n \to X_n$ satisfying $\alpha_n(u, F_n(u, y)) \leq \beta_n(fu, y)$. The category is symmetric monoidal closed, with products and weak coproducts, and for every lineale it carries a linear exponential comonad, the multisets of counterexamples being taken degree by degree (Section 5); so it is a model of intuitionistic linear logic with its modality.

Vector spaces do not seem to give the picture we want. Over an additive base the multiplication of any internal lineale is addition and its order is equality, so every morphism condition is an equation and the tensor of the construction is its product (Section 6); reading relations as subspaces instead does not give a better category (Section 6). Over **FinVect** there is moreover no internal hom of the form used here, since the cofree coalgebras it needs are not finite-dimensional. Over all of **Vect** the internal hom exists, and $\mathsf{Dial}_\Omega(\mathbf{Vect})$ turns out to be cartesian closed rather than a model of linear logic with a non-trivial modality. We record the calculation because it places the obstruction in the additivity of the base, not in its dimension.

2 Dial over Set

Before moving off a cartesian base we recall the construction over **Set**, in the form used throughout the note. Over sets a relation between witnesses and counterexamples can be given either as a subset of $U \times X$ or as its characteristic map into the booleans. The second reading generalizes by replacing the booleans with a lineale, a poset of truth values with a residuated commutative monoid structure [8]. This is the form taken in the companion note [7], and the one we keep over a monoidal base, for the reasons given in Remark 3.2. The formulas recalled here, for the tensor, the internal hom, products, weak coproducts and the modality of $\mathsf{Dial}_\Omega(\mathbf{Set})$, are the ones that Sections 3 and 5 generalize, so each construction there can be checked against its value over **Set**.

Definition 2.1 (Lineale [8])**.** A *lineale* is a poset $(\Omega, \leq)$ with a commutative monoid structure $(\circ, e)$, monotone in each argument, such that for all a, b there is a largest c with $a \circ c \leq b$, written $a \multimap b$.

A lineale need not be complete. Every commutative quantale is a lineale, as is every Heyting algebra with $\circ = \wedge$, the booleans among them, and the natural numbers with truncated subtraction, the lineale of [3]. The effect of varying the lineale is studied in the poset-valued models of Andrea Schalk and myself [4].

Over **Set** the objects of $\mathsf{Dial}_\Omega(\mathbf{Set})$ are triples (U, X, α) with $\alpha \colon U \times X \to \Omega$, and a morphism $(f, F) \colon (U, X, \alpha) \to (V, Y, \beta)$ is a pair of maps $f \colon U \to V$ and $F \colon U \times Y \to X$ making the square

$$\begin{array}{ccc} U \times Y & \xrightarrow{\langle \pi_1, F \rangle} & U \times X \\ {\scriptstyle f\times Y}\downarrow & \leq & \downarrow{\scriptstyle \alpha} \\ V \times Y & \xrightarrow[\beta]{} & \Omega \end{array}$$

commute up to the order of Ω, that is, $\alpha(u, F(u,y)) \leq \beta(fu, y)$. The structure of $\mathsf{Dial}_\Omega(\mathbf{Set})$ [2] is given by a tensor that is a product in both coordinates and an internal hom that carries the witness into the backward map,

$$\begin{aligned} A \otimes B &= \big(U \times V,\ X \times Y,\ \alpha \otimes \beta\big), \\ (\alpha \otimes \beta)\big((u,v),(x,y)\big) &= \alpha(u,x) \circ \beta(v,y), \\ [A,B] &= \big(V^U \times X^{U\times Y},\ U \times Y,\ \alpha \multimap \beta\big), \\ (\alpha \multimap \beta)\big((f,F),(u,y)\big) &= \alpha\big(u, F(u,y)\big) \multimap \beta(fu,y), \end{aligned}$$

with tensor unit $(1,1,e)$. Products are $A \,\&\, B = (U \times V, X + Y, \alpha \,\&\, \beta)$, with the relation given by cases, and the terminal object is $(1,0,!)$, where $!$ is the unique map out of the empty set. There are only weak coproducts, $(U+V, X^U \times Y^V, \alpha \oplus \beta)$, whose relation evaluates the counterexample function at the witness [2, pp. 13–14], and the initial object is $(0,1,!)$. The modality [2] is $!(U,X,\alpha) = (U, X^*, \alpha^*)$, with X^* the free commutative monoid on X and α^* the product in Ω over a multiset.

3 Dial over a monoidal base

We now replace **Set**, or more generally a cartesian closed base, by a symmetric monoidal closed category $\mathcal{V}$ with finite products and coproducts, and check that each piece of structure recalled in Section 2 can still be built. Three things have to be supplied that a cartesian base provides without being asked. Truth values must be carried by an object of the base, so we need a lineale internal to it. Witnesses must be copied and discarded, since the backward map of a dialectica morphism uses the witness both to produce a counterexample and to evaluate the relation, so witnesses have to be comonoids. Composition and the unit laws then need forward maps to respect this structure, so forward maps have to be comonoid morphisms. With these in place we show that the construction gives a category, a symmetric monoidal structure, products, weak coproducts and an internal hom under a hypothesis on comonoids. Over **Set** each of these reduces to the corresponding formula of Section 2.

We write $(\mathcal{V}, \otimes, I, \multimap)$ for the base, keeping $\mathcal{C}$ for cartesian closed categories. Since $\otimes$ is a left adjoint in each variable, it preserves coproducts, so distributivity never has to be assumed. What must be assumed is the truth-value object.

Definition 3.1 (Internal lineale)**.** An *internal lineale* in $\mathcal{V}$ is an object Ω with

1. a commutative monoid structure $\circ\colon \Omega \times \Omega \to \Omega$, $e\colon 1 \to \Omega$, monotone in each argument;
2. an internal partial order $\leq \rightarrowtail \Omega \times \Omega$, reflexive, transitive and antisymmetric;
3. a map $\multimap\colon \Omega \times \Omega \to \Omega$ satisfying the residuation adjunction internally.

Here $p \leq q$, for parallel maps $p, q\colon D \to \Omega$, means that $\langle p, q\rangle$ factors through $\leq$, and reflexivity, transitivity, antisymmetry, monotonicity and residuation are required of all parallel maps $p, q, r\colon D \to \Omega$, so that no limits beyond finite products are needed.

Both the multiplication and the residuation are maps out of $\Omega \times \Omega$, because in the tensor and in the internal hom of either construction they combine two truth values computed from the same object, as $\circ \circ \langle p, q\rangle$ and $\multimap \circ \langle p, q\rangle$ for parallel $p, q\colon D \to \Omega$. The pairing needs only products in $\mathcal{V}$, and no comonoid structure on D.

Remark 3.2. This is the hypothesis that decides which bases are available, and it is more demanding than the monoidal ones, which most familiar categories satisfy. It asks for an order carried by an object of the base, together with a genuinely binary residuation on it. The reading

of a relation as a map into Ω is what makes the demand; taking relations as subobjects instead changes the question, and changes it in a way that matters for bases of a linear-algebraic kind, where the order one wants is supplied by a closure operator rather than by the base's own structure. That reading is taken up in Section 6.

Definition 3.3. A *comonoid* in $\mathcal{V}$ is $(U, \delta_U \colon U \to U \otimes U, \varepsilon_U \colon U \to I)$, coassociative, counital and cocommutative; $\mathbf{Comon}(\mathcal{V})$ is the category of these and the maps commuting with the structure.

Proposition 3.4 (Folklore)**.** $\mathbf{Comon}(\mathcal{V})$ *has finite products, given by* $\otimes$*, with terminal object* I*; that is,* $\mathbf{Comon}(\mathcal{V})$ *is cartesian, and the forgetful functor to* $\mathcal{V}$ *is strong monoidal.*

Definition 3.5. $\mathsf{Dial}_\Omega(\mathcal{V})$ has objects (U, X, α) with U a cocommutative comonoid, X an object carrying no structure, and $\alpha \colon U \otimes X \to \Omega$. A morphism to (V, Y, β) is a comonoid morphism $f \colon U \to V$ together with $F \colon U \otimes Y \to X$, subject to

$$\begin{array}{ccccc} U \otimes Y & \xrightarrow{\delta_U \otimes Y} & U \otimes U \otimes Y & \xrightarrow{U \otimes F} & U \otimes X \\ {\scriptstyle f \otimes Y}\downarrow & & \leq & & \downarrow{\scriptstyle \alpha} \\ V \otimes Y & & \xrightarrow[\beta]{\qquad\qquad\qquad\qquad} & & \Omega \end{array}$$

that is, the composite through α lies below the composite through β; in elements, $\alpha(u, F(u, y)) \leq \beta(fu, y)$.

Proposition 3.6. $\mathsf{Dial}_\Omega(\mathcal{V})$ *is a category.*

Proof. For the identity on (U, X, α) take $f = \mathrm{id}_U$ and $F = \lambda \circ (\varepsilon_U \otimes \mathrm{id}_X) \colon U \otimes X \to I \otimes X \cong X$. The left-hand side of the condition is then $\alpha \circ (\mathrm{id}_U \otimes \lambda) \circ (\mathrm{id}_U \otimes \varepsilon_U \otimes \mathrm{id}_X) \circ (\delta_U \otimes \mathrm{id}_X) = \alpha$ by the counit law, and the right-hand side is α; reflexivity gives the condition. The counit is playing the role of a projection.

For composition, let $(f, F) \colon A \to B$ and $(g, G) \colon B \to C = (W, Z, \gamma)$. Take $h = g \circ f$ and

$$H \;=\; F \circ (\mathrm{id}_U \otimes G) \circ (\mathrm{id}_U \otimes f \otimes \mathrm{id}_Z) \circ (\delta_U \otimes \mathrm{id}_Z) \;:\; U \otimes Z \to X,$$

that is: duplicate the witness, push one copy forward along f and feed it with the counterexample to G, then feed the resulting counterexample, together with the other copy, to F. For the condition, apply the first inequality with the counterexample $G \circ (f \otimes \mathrm{id}_Z)$ and then the second, using coassociativity of δ_U to supply the three copies of the witness and transitivity of $\leq$ to join the two steps.

The unit laws and associativity are where the forward map has to be a comonoid morphism. Composing (f, F) with the identity of B gives, up to the unitor, the backward part $F \circ (\mathrm{id}_U \otimes (\varepsilon_V \circ f) \otimes \mathrm{id}_Y) \circ (\delta_U \otimes \mathrm{id}_Y)$, which is F because $\varepsilon_V \circ f = \varepsilon_U$ and U is counital; composing with the identity of A needs only counitality of U. In the two bracketings of a triple composite the witness is used three times, but in one of them f is applied separately to two copies and in the other f is applied once and its value copied, so the two backward parts agree because $(f \otimes f) \circ \delta_U = \delta_V \circ f$, coassociativity doing the rest. □

Remark 3.7. Over a cartesian base every map is a comonoid morphism for the diagonal structure, so the condition on f costs nothing and has never needed stating; over **GrSet** it is likewise automatic, by Proposition 4.4. Only the forward map is constrained: the backward map lands in X, which carries no structure.

3.1 Tensor

The tensor combines witnesses by the tensor of $\mathcal{V}$ and counterexamples by its product, and its relation multiplies the two truth values in the lineale. Each factor's relation must see only its own witness, so the unused one is discarded with a counit, and this is the first place the comonoid structure on witnesses is used.

Definition 3.8. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ set $A \otimes B = (U \otimes V,\ X \times Y,\ \alpha \otimes \beta)$, where $U \otimes V$ carries the comonoid structure of Proposition 3.4 and $\alpha \otimes \beta$ is

$$(U \otimes V) \otimes (X \times Y) \longrightarrow \big(U \otimes V \otimes X\big) \times \big(U \otimes V \otimes Y\big) \longrightarrow (U \otimes X) \times (V \otimes Y) \xrightarrow{\alpha \times \beta} \Omega \times \Omega \xrightarrow{\circ} \Omega,$$

the first map pairing the two projections of $X \times Y$ and the second discarding the unused witness with a counit. In elements, $(\alpha \otimes \beta)((u, v), (x, y)) = \alpha(u, x) \circ \beta(v, y)$. The unit is $\mathbf{1} = (I, 1, e)$.

As in [1], the witnesses are combined by the tensor of $\mathcal{V}$ and the counterexamples by its product; over **Set** Definition 3.8 is the familiar $(U \times V, X \times Y, \alpha \otimes \beta)$ of Section 2.

Theorem 3.9. $(\mathsf{Dial}_\Omega(\mathcal{V}), \otimes, \mathbf{1})$ *is a symmetric monoidal category.*

Proof. On morphisms, given $(f, F)\colon A \to A'$ and $(g, G)\colon B \to B'$, take the forward part $f \otimes g$ and the backward part $(U \otimes V) \otimes (X' \times Y') \to X \times Y$ that pairs F, applied after discarding v and projecting to X', with G, applied after discarding u and projecting to Y'. In elements it sends $((u, v), (x', y'))$ to $(F(u, x'), G(v, y'))$, so the condition reads

$$\alpha(u, F(u, x')) \circ \beta(v, G(v, y')) \ \leq\ \alpha'(fu, x') \circ \beta'(gv, y'),$$

which follows from the two given inequalities and monotonicity of $\circ$ in each argument.

The structural isomorphisms are those of $\mathcal{V}$ on witnesses and those of the cartesian structure on counterexamples, with backward parts supplied by the counits as in Proposition 3.6. Their conditions hold as equations, by associativity, commutativity and unitality of $(\Omega, \circ, e)$, and they satisfy the coherence conditions of a symmetric monoidal category by Proposition 3.12 below. □

3.2 Coherence

The proof of Theorem 3.9 leaves the coherence conditions to this subsection. The backward parts of the structural maps never read the witness, and for morphisms of this kind coherence in $\mathsf{Dial}_\Omega(\mathcal{V})$ reduces to coherence in $\mathcal{V}$. A diagram in $\mathsf{Dial}_\Omega(\mathcal{V})$ commutes if and only if the diagrams of forward parts and of backward parts both commute in $\mathcal{V}$, since the inequality of Definition 3.5 is a property of a pair and not structure on it.

Definition 3.10. A morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ is *inert* if its backward part discards the witness, that is if $F = s \circ \lambda \circ (\varepsilon_U \otimes \mathrm{id}_Y)$ for some $s\colon Y \to X$ in $\mathcal{V}$. We then write the morphism as $(f \mid s)$.

Lemma 3.11. *Let $(f, F)\colon A \to B$ and $(g, G)\colon B \to C$.*

1. *If $(g, G) = (g \mid s)$ is inert, the composite has backward part $F \circ (\mathrm{id}_U \otimes s)$.*
2. *If $(f, F) = (f \mid s)$ is inert, the composite has backward part $s \circ G \circ (f \otimes \mathrm{id}_Z)$.*
3. *If both are inert, the composite is $(g \circ f \mid s \circ s')$.*

Proof. In each case the counit in the inert backward part meets a comultiplication and cancels by the counit law. □

Proposition 3.12. *The associator, the unitors and the symmetry of Theorem 3.9 are inert, their forward parts being the structural isomorphisms of $\mathcal{V}$ on witnesses and their backward parts the inverse structural isomorphisms on counterexamples. Consequently the pentagon, the triangle and the hexagon in $\mathsf{Dial}_\Omega(\mathcal{V})$ commute, and each structural isomorphism is natural.*

Proof. By Lemma 3.11(iii) a composite of structural maps is inert with components the composites of the two families, so each coherence diagram in $\mathsf{Dial}_\Omega(\mathcal{V})$ is the corresponding diagram of $\mathcal{V}$ read on witnesses together with the one read on counterexamples in the opposite direction, and these commute. Naturality against an arbitrary (f, F) reduces by Lemma 3.11(i) and (ii) to naturality in $\mathcal{V}$. Each structural map satisfies the inequality of Definition 3.5 as an equation, by associativity, unitality and commutativity of $(\Omega, \circ, e)$. □

So the order on Ω plays no part in coherence. What is not automatic over a monoidal base is that $\otimes$ is a bifunctor on morphisms that are not inert, and that is where the comonoid laws and the monotonicity of $\circ$ are used, in the proof of Theorem 3.9.

3.3 Products

As over **Set**, products pair the witnesses and take the coproduct of the counterexamples, the relation being given by cases. The tensor of $\mathcal{V}$ serves on witnesses here as well, because forward maps are comonoid morphisms and $\otimes$ is the product in $\mathbf{Comon}(\mathcal{V})$ (Proposition 3.4): a comonoid morphism $W \to U \otimes V$ is determined by its composites with the projections $p_1 = \rho \circ (\mathrm{id}_U \otimes \varepsilon_V)$ and $p_2 = \lambda \circ (\varepsilon_U \otimes \mathrm{id}_V)$. No hypothesis on the products of $\mathcal{V}$ is needed for this.

Definition 3.13. Set

$$A \,\&\, B \;=\; \big(U \otimes V,\;\; X + Y,\;\; \alpha \,\&\, \beta\big),$$

where $U \otimes V$ carries the comonoid structure of Proposition 3.4 and, using $(U \otimes V) \otimes (X + Y) \cong \big((U \otimes V) \otimes X\big) + \big((U \otimes V) \otimes Y\big)$, the relation is the copairing of $\alpha \circ (p_1 \otimes \mathrm{id}_X)$ and $\beta \circ (p_2 \otimes \mathrm{id}_Y)$.

Theorem 3.14. *The object $A \,\&\, B$ is the cartesian product of A and B in $\mathsf{Dial}_\Omega(\mathcal{V})$, and $\top = (I, 0, !)$ is terminal.*

Proof. The first projection is $(p_1, \iota_1 \circ \lambda \circ (\varepsilon_{U \otimes V} \otimes \mathrm{id}_X))$; its forward part is a comonoid morphism by Proposition 3.4, and its condition holds as an equation by construction of the relation and the counit law. Similarly the second. Given $(f, F) \colon C \to A$ and $(g, G) \colon C \to B$ with $C = (W, Z, \gamma)$, the mediating morphism has forward part $(f \otimes g) \circ \delta_W$, the unique comonoid morphism with the required composites, and backward part the copairing $[F, G]$ under $W \otimes (X + Y) \cong (W \otimes X) + (W \otimes Y)$, unique because a map out of a coproduct is determined by its restrictions. The condition splits into the two summands and holds there by hypothesis, since $p_1 \circ (f \otimes g) \circ \delta_W = f$ and $p_2 \circ (f \otimes g) \circ \delta_W = g$. Terminality of $\top$ holds because ε_W is the unique comonoid morphism $W \to I$, the backward part being the unique map out of 0. □

3.4 Weak coproducts

As over **Set**, the construction has only weak coproducts: the counterexamples of $A \oplus B$ are pairs of functions, and the relation evaluates them at the witness. Over a monoidal base these function spaces are internal homs of $\mathcal{V}$, and evaluating at the witness while keeping it for the relation needs a copy of it, so here the comonoid structure enters the relation itself.

Definition 3.15. Since $\otimes$ preserves coproducts, $U + V$ is a comonoid, with counit $[\varepsilon_U, \varepsilon_V]$ and comultiplication $U + V \to (U \otimes U) + (V \otimes V) \to (U + V) \otimes (U + V)$; for this structure the injections, and copairings of comonoid morphisms, are comonoid morphisms. Set

$$A \oplus B \;=\; \big(U + V,\; [U, X] \times [V, Y],\; \alpha \oplus \beta\big),$$

where, under $(U+V)\otimes W \cong (U\otimes W)+(V\otimes W)$, the relation on the first summand is

$$U\otimes([U,X]\times[V,Y]) \longrightarrow U\otimes[U,X] \xrightarrow{\delta_U\otimes\mathrm{id}} U\otimes U\otimes[U,X] \xrightarrow{\mathrm{id}\otimes\mathrm{ev}} U\otimes X \xrightarrow{\alpha} \Omega,$$

and similarly on the second. In elements, $(\alpha\oplus\beta)(\iota_1 u,(\varphi,\psi)) = \alpha(u,\varphi u)$.

Proposition 3.16. *The injection ι_1, with forward part the coproduct injection and backward part $(u,(\varphi,\psi))\mapsto\varphi u$, and likewise ι_2, are morphisms of $\mathsf{Dial}_\Omega(\mathcal{V})$, and their conditions hold as equations. Every pair $(f,F)\colon A\to C$ and $(g,G)\colon B\to C$ has a mediating morphism $A\oplus B\to C$, which is not unique in general, so $A\oplus B$ is a weak coproduct.*

Proof. Let $C=(W,Z,\gamma)$. The forward part is $[f,g]\colon U+V\to W$, a comonoid morphism by the choice of comonoid structure on $U+V$. For the backward part $(U+V)\otimes Z\to[U,X]\times[V,Y]$ take, on each summand, the map that discards the witness with the counit and sends z to $(\hat{F}z,\hat{G}z)$, where $\hat{F}\colon Z\to[U,X]$ and $\hat{G}\colon Z\to[V,Y]$ are the transposes of F and G. At $\iota_1 u$ the condition reads $\alpha(u,F(u,z))\le\gamma(fu,z)$, which is the condition for (f,F); similarly at $\iota_2 v$. Composing with ι_1 evaluates $\hat{F}z$ at u and so returns (f,F). Uniqueness fails because on the summand $U\otimes Z$ the backward part may also read u: any map sending (u,z) to a pair (φ,ψ) with $\varphi u=F(u,z)$ serves equally well. □

The nullary case is a genuine initial object: since $\otimes$ preserves colimits, $0\otimes W\cong 0$, so 0 is a comonoid, initial in $\mathbf{Comon}(\mathcal{V})$, and $\mathbf{0}=(0,1,!)$ is initial in $\mathsf{Dial}_\Omega(\mathcal{V})$, every backward map out of $0\otimes Y\cong 0$ being unique; as over $\mathbf{Set}$, every $(0,Z,!)$ is initial, and these objects are all isomorphic. Over $\mathbf{Set}$ the binary case is the weak coproduct $(U+V, X^U\times Y^V, \alpha\oplus\beta)$ of the thesis, recalled in Section 2.

3.5 The internal hom

The internal hom is the one place where moving off a cartesian base asks for more than comonoids. Its witnesses are pairs of a forward and a backward map, and they must themselves form a comonoid, which an object of $\mathcal{V}$ need not be. We take the cofree comonoid on the object of such pairs, and closure then needs every map between comonoids to be a comonoid morphism.

Definition 3.17. Assume that $\mathcal{V}$ has cofree comonoids, that is, that the forgetful functor from $\mathbf{Comon}(\mathcal{V})$ to $\mathcal{V}$ has a right adjoint $\sharp$. Put

$$[A,B] \;=\; \Big(\sharp\big([U,V]\times[U\otimes Y,X]\big),\; U\otimes Y,\; \gamma\Big),$$

where γ is $\multimap\circ\langle p,q\rangle$ with p the composite that evaluates the second factor at (u,y) and applies α, using δ_U twice so that u is available to the backward map and to the relation, and q the composite that evaluates the first factor at u and applies β.

Theorem 3.18. *Suppose in addition that every map of $\mathcal{V}$ between comonoids is a comonoid morphism. Then $\mathsf{Dial}_\Omega(\mathcal{V})(A\otimes B,C)\cong\mathsf{Dial}_\Omega(\mathcal{V})(A,[B,C])$, naturally, so $\mathsf{Dial}_\Omega(\mathcal{V})$ is symmetric monoidal closed.*

Proof sketch. The forward part of a morphism $A\to[B,C]$ is a comonoid morphism $U_A\to\sharp\big([U_B,U_C]\times[U_B\otimes Y_C,X_C]\big)$, which by cofreeness is the same as a pair of maps $U_A\to[U_B,U_C]$ and $U_A\to[U_B\otimes Y_C,X_C]$, that is, by closedness of $\mathcal{V}$, a map $U_A\otimes U_B\to U_C$ and a map $U_A\otimes U_B\otimes Y_C\to X_C$. The backward part is a map $U_A\otimes U_B\otimes Y_C\to X_A$. By the added hypothesis the map into U_C is a comonoid morphism, and since the counterexamples of $A\otimes B$ are the product $X_A\times X_B$, the two backward maps together are exactly one map $U_A\otimes U_B\otimes Y_C\to X_A\times X_B$. So the data of a morphism $A\to[B,C]$ are exactly those of a morphism $A\otimes B\to C$. The two conditions,

$$\alpha(u,F_1)\circ\beta(v,F_2)\le\gamma(f(u,v),z)\quad\text{and}\quad\alpha(u,F_1)\le\beta(v,F_2)\multimap\gamma(f(u,v),z),$$

with F_1 and F_2 evaluated at (u, v, z), correspond by the residuation adjunction of Ω. Naturality is inherited from that of the adjunctions used. □

Remark 3.19. The hypothesis holds over a cartesian base and over **GrSet**, by Proposition 4.4, and it implies that maps out of a comonoid cannot tell $\sharp Z$ from Z, which is what the proof uses. Without it the forward part of a morphism $A \otimes B \to C$ must be a comonoid morphism while its transpose need only be a map into $[U_B, U_C]$, and the witness object of $[A, B]$ would have to be built from comonoid morphisms instead; we do not pursue that here.

3.6 Summary

Table 1 collects the structure of $\mathsf{Dial}_\Omega(\mathcal{V})$ constructed in this section, with the corresponding objects of $\mathsf{Dial}_\Omega(\mathbf{Set})$ from Section 2 alongside. Witnesses are combined by the tensor of $\mathcal{V}$ throughout and the products of $\mathcal{V}$ appear only on counterexamples; the further hypotheses, cofree comonoids and maps between comonoids being comonoid morphisms, are needed only for the internal hom.

	over $\mathcal{V}$	relation, in elements	over **Set**
unit $\mathbf{1}$	$(I,\ 1,\ e)$	e	$(1, 1, e)$
$A \otimes B$	$(U \otimes V,\ X \times Y)$	$\alpha(u, x) \circ \beta(v, y)$	$(U \times V,\ X \times Y)$
$[A, B]$	$(\sharp([U, V] \times [U \otimes Y, X]),\ U \otimes Y)$	$\alpha(u, F(u, y)) \multimap \beta(fu, y)$	$(V^U \times X^{U \times Y},\ U \times Y)$
$\top$	$(I,\ 0)$	!	$(1, 0)$
$\mathbf{0}$	$(0,\ 1)$	!	$(0, 1)$
$A \,\&\, B$	$(U \otimes V,\ X + Y)$	by cases	$(U \times V,\ X + Y)$
$A \oplus B$ (weak)	$(U + V,\ [U, X] \times [V, Y])$	$\alpha(u, \varphi u),\ \beta(v, \psi v)$	$(U + V,\ X^U \times Y^V)$

Table 1: The structure of $\mathsf{Dial}_\Omega(\mathcal{V})$, with witnesses and counterexamples, and its value over **Set**.

4 Graded sets

The construction of Section 3 needs a concrete example: a base that is symmetric monoidal closed but not cartesian, that satisfies the hypotheses of that section, and that mathematicians already use for reasons of their own rather than one built to fit the theorem. Graded sets are such a base. A graded set is a set whose elements each have a size, and the Cauchy product[1] combines two of them by pairing elements and adding their sizes. Graded sets with finitely many elements of each size are the combinatorial classes of symbolic combinatorics [25], where the Cauchy product corresponds to the product of generating functions.

We describe the category, check that a constant lineale is internal to it and that its comonoids are plain sets, and then read off what dialectica objects and morphisms over it are. Since every map between its comonoids is a comonoid morphism, the conditions on witnesses and forward maps of Section 3 hold automatically over **GrSet**, and what differs from the cartesian case is carried by the counterexamples. The answer is concrete enough to compute with, and it is the base over which Section 5 builds the modality.

Definition 4.1. A *graded set* is a family $F = (F_n)_{n \in \mathbb{N}}$ of sets, and a morphism $F \to G$ is a family of functions $F_n \to G_n$. Write **GrSet** for the category of these. The *Cauchy product* and

[1]The name comes from Cauchy's product of power series, in which the coefficient of z^n in $\left(\sum_i a_i z^i\right)\left(\sum_j b_j z^j\right)$ is $\sum_{i+j=n} a_i b_j$. The Cauchy product of graded sets, $(X \otimes Y)_n = \coprod_{i+j=n} X_i \times Y_j$, is the same formula one level up: when every X_i and Y_j is finite, $|(X \otimes Y)_n| = \sum_{i+j=n} |X_i|\,|Y_j|$, so the generating function of $X \otimes Y$ is the product of those of X and Y. See [25, Chapter I], where it is the product construction on combinatorial classes.

its internal hom are

$$(F \otimes G)_n = \coprod_{i+j=n} F_i \times G_j, \qquad [F,G]_n = \prod_i \mathbf{Set}(F_i,\, G_{n+i}),$$

and the unit I is a one-element set in degree 0 and empty elsewhere.

Proposition 4.2. *The category* $(\mathbf{GrSet}, \otimes, I)$ *is symmetric monoidal closed, with products and coproducts computed degreewise,* $(F \times G)_n = F_n \times G_n$ *and* $(F + G)_n = F_n + G_n$*. It is not cartesian.*

Proof. Associativity, symmetry and unitality of $\otimes$ are those of $\times$ and $+$ in **Set** together with those of addition in $\mathbb{N}$. For closure, a morphism $F \otimes G \to H$ is a family of functions $F_i \times G_j \to H_{i+j}$, which is the same as a family $F_n \to \prod_j \mathbf{Set}(G_j, H_{n+j}) = [G,H]_n$. The base is not cartesian since the unit is not terminal: the terminal graded set is a one-element set in every degree. □

This is the Day convolution [6] on presheaves over the discrete monoidal category $(\mathbb{N}, +)$, but nothing from that theory is used here. Equivalently, **GrSet** is the slice $\mathbf{Set}/\mathbb{N}$ of sets equipped with a size function, and the Cauchy product is the cartesian product with sizes added.

Proposition 4.3. *For any lineale* Ω*, the constant graded set with* Ω *in every degree, again written* Ω*, is an internal lineale in* **GrSet***.*

Proof. Products and subobjects are degreewise, so $\leq$, $\circ$, e and $\multimap$ can be taken to be those of Ω in each degree, and the laws hold degreewise because they hold in Ω. □

Proposition 4.4. *In* **GrSet** *the comonoids are exactly the graded sets concentrated in degree* 0*, so* $\mathbf{Comon}(\mathbf{GrSet}) \simeq \mathbf{Set}$*; every map between such is a comonoid morphism; cofree comonoids exist,* $\sharp Z = Z_0$*; the hypothesis of Theorem 3.18 holds; and on degree-*0 *objects* $\otimes$ *and* $\times$ *agree.*

Proof. The unit is concentrated in degree 0, so a counit $\varepsilon\colon F \to I$ forces $F_n = \emptyset$ for $n > 0$; conversely a degree-0 object is uniquely a comonoid by the diagonal, since there $(F \otimes G)_0 = F_0 \times G_0$ and $(F \otimes G)_n = \emptyset$ for $n > 0$, so $\otimes$ and $\times$ agree on such objects, and $\mathbf{Comon}(\mathbf{GrSet}) \simeq \mathbf{Set}$ is cartesian in the same way **Set** is. For cofreeness, $\mathbf{GrSet}(iU, Z) = \prod_n \mathbf{Set}((iU)_n, Z_n) = \mathbf{Set}(U, Z_0)$, since $(iU)_n = \emptyset$ for $n > 0$; the same computation gives $\mathbf{GrSet}(iU, \sharp Z) = \mathbf{Set}(U, Z_0) = \mathbf{GrSet}(iU, Z)$, which is the hypothesis of Theorem 3.18. Degree-0 objects are closed under coproducts pointwise. □

Corollary 4.5. $\mathsf{Dial}_\Omega(\mathbf{GrSet})$ *is a symmetric monoidal closed category with finite products and weak coproducts, for every lineale* Ω*. Concretely: an object is a set* U*, a graded set* X *and functions* $\alpha_j\colon U \times X_j \to \Omega$*; a morphism is* $f\colon U \to V$ *together with* $F_j\colon U \times Y_j \to X_j$ *satisfying* $\alpha_j(u, F_j(u,y)) \leq \beta_j(fu, y)$*; the tensor is* $(U \times V, X \times Y, \alpha \otimes \beta)$*, with the counterexamples combined degreewise; and*

$$\sharp\big([U \otimes Y, X]\big) \;=\; \prod_j \mathbf{Set}(U \times Y_j, X_j),$$

which is precisely the set of backward maps. The witnesses are plain sets and the counterexamples are graded, and this is what makes the internal hom computable.

5 The modality

The structure of Section 3 models the multiplicative and additive fragment of intuitionistic linear logic; to model the whole of it we need a modality, a linear exponential comonad whose co-Kleisli category is cartesian closed and so interprets intuitionistic implication. Over **Set** the modality of Section 2 leaves the witnesses alone and replaces the counterexamples by finite multisets of them,

combining their truth values by the multiplication of the lineale. Obtaining a modality over a base that is merely symmetric monoidal closed, rather than cartesian closed, is the question left open in the conclusions of [3]. We do not answer it for a general $\mathcal{V}$, but over **GrSet** the construction goes through.

The reason is the shape of the tensor. Since the counterexamples of a tensor are combined by the product of the base, the monoid wanted on counterexamples is a monoid for the product, and in **GrSet** products are computed degree by degree. So the multisets are taken in each degree separately, while the witnesses, which are plain sets, are untouched. We define the modality in this way, check that it is a comonad satisfying the Girard isomorphisms, that each $!A$ is a commutative comonoid, and that the two structures are compatible; these are the conditions in the Bierman–Hyland–Schalk characterisation of a linear exponential comonad.

Proposition 5.1. *The free commutative monoid on X for the product of* **GrSet** *is*

$$(X^*)_d \;=\; \big\{\, \textit{finite multisets of elements of } X_d \,\big\},$$

with the empty multiset in every degree.

Definition 5.2. For $A = (U, X, \alpha)$ in $\mathsf{Dial}_\Omega(\mathbf{GrSet})$ put $!A = (U, X^*, \alpha^*)$ with, in each degree d and for a multiset $\{x_1, \ldots, x_k\}$ of elements of X_d,

$$\alpha^*_d\big(u, \{x_1, \ldots, x_k\}\big) \;=\; \alpha_d(u, x_1) \circ \cdots \circ \alpha_d(u, x_k),$$

the empty multiset going to the unit e of the lineale.

This is the formula of [3], with the lineale's multiplication rather than its meet. Over $\Omega = 2$ the two agree, which is why the formula is often glossed with a conjunction; over the lineale $\mathbb{N}$ used for Petri nets it is addition, so $!\alpha$ records total multiplicity. With a meet, contraction would need $a \wedge b \leq a \circ b$ and the empty multiset would force $e = \top$; with $\circ$ neither condition arises.

Proposition 5.3. *The construction $!$ is a functor. Given $(f, F)\colon A \to B$, put $!(f, F) = (f, F^*)$, where*

$$F^*\big(u, \{y_1, \ldots, y_k\}\big) = \big\{F(u, y_1), \ldots, F(u, y_k)\big\},$$

degree by degree. The required inequality, $\bigcirc_i \alpha(u, F(u, y_i)) \leq \bigcirc_i \beta(fu, y_i)$, follows from the given inequality for each i and monotonicity of $\circ$, with no hypothesis on Ω beyond Definition 2.1.

Proposition 5.4. *The functor $!$ is a comonad. Its counit has forward part id_U and backward part $(u, x) \mapsto \{x\}$, correct because $\alpha^*(u, \{x\}) = \alpha(u, x)$; its comultiplication has forward part id_U and backward part the union of a multiset of multisets, correct because $\circ$ is associative and commutative, so both sides of the inequality are the product over the same underlying family. The comonad laws are the monoid laws of X^*.*

Proposition 5.5 (Girard isomorphisms)**.** $!(A \,\&\, B) \cong !A \otimes !B$, *and* $!\top \cong \mathbf{1}$.

Proof. The free commutative monoid functor is a left adjoint, so it carries coproducts to coproducts, and the coproduct of commutative monoids for the cartesian structure is the product; hence $(X + Y)^* \cong X^* \times Y^*$. Concretely, a multiset of elements of $X_d + Y_d$ is the same thing as a pair of multisets, one from X_d and one from Y_d. The relations agree on the nose: the product over a multiset from $X + Y$ splits, by commutativity of $\circ$, into the X-part times the Y-part, which is the relation of $!A \otimes !B$, the counits discarding the unused witness on each side. At the level of witnesses the isomorphism is the identity of $U \otimes V$, which is the witness object of both sides by Definition 3.13. For the unit, $0^* = 1$ carries only the empty multiset in each degree, with relation e, which is $\mathbf{1}$. □

Proposition 5.6. *$!A$ is a commutative comonoid in $\mathsf{Dial}_\Omega(\mathbf{GrSet})$, with weakening given by the unit $1 \to X^*$ of the monoid and contraction by its multiplication $X^* \times X^* \to X^*$. The two inequalities hold as equations: $\alpha^*(u, \varnothing) = e$ and $\alpha^*(u, m \cup m') = \alpha^*(u, m) \circ \alpha^*(u, m')$.*

5.1 The linear exponential comonad

Everything above is a statement about the free commutative monoid X^* transported through α^*. The following lemma makes that explicit and does all the remaining work.

Lemma 5.7. *For each degree d and each $u \in U$ the map $\alpha^*_d(u,-)\colon (X^*_d, \cup, \varnothing) \to (\Omega, \circ, e)$ is a homomorphism of commutative monoids, and it is the unique one extending $\alpha_d(u,-)$ along the insertion of generators $X_d \to X^*_d$.*

Proof. Definition 5.2 sets $\alpha^*_d(u,m)$ to be the product in Ω over the multiset m of elements of X_d, so it takes $\varnothing$ to e and $m \cup m'$ to $\alpha^*_d(u,m) \circ \alpha^*_d(u,m')$; commutativity and associativity of $\circ$ make this well defined on multisets. Uniqueness is the universal property of X^*_d. □

The compatibility condition for a linear exponential comonad is that the comonoid structure carried by $!A$ be the one induced, through the Girard isomorphisms, by the diagonal $A \to A \,\&\, A$ and the unique map $A \to \top$. We check it.

Proposition 5.8. *Let $\Delta = (\delta_U, [\mathrm{id}_X, \mathrm{id}_X])\colon A \to A \,\&\, A$ be the diagonal and $s\colon !(A \,\&\, A) \xrightarrow{\cong} !A \otimes !A$ the Girard isomorphism of Proposition 5.5. Then*

$$s \circ !\Delta \;=\; d_A, \qquad (! \text{ of the unique map } A \to \top) \text{ followed by } !\top \cong \mathbf{1} \;=\; e_A,$$

where (e_A, d_A) is the comonoid structure of Proposition 5.6. Consequently δ_A is a comonoid morphism, e_A and d_A are coalgebra morphisms, and every map of free coalgebras is a map of comonoids.

Proof. Forward parts first. The forward part of $!\Delta$ is $\delta_U\colon U \to U \otimes U$ and that of s is the identity, by Proposition 5.5; the forward part of d_A is δ_U. They agree.

Backward parts. Write $m, m' \in X^*$. The backward part of s at (m, m') is $\iota_1 m \cup \iota_2 m' \in (X + X)^*$, tagging each multiset with its side; the backward part of $!\Delta$ is $M \mapsto$ the image of M under the codiagonal $[\mathrm{id}_X, \mathrm{id}_X]$, which erases tags. Composing,

$$(m, m') \;\longmapsto\; [\mathrm{id}, \mathrm{id}]\big(\iota_1 m \cup \iota_2 m'\big) \;=\; m \cup m',$$

because the image of a union of multisets is the union of the images and erasing a tag is the identity on the underlying element. That is exactly the backward part of d_A, namely the multiplication of X^*. The nullary case is the same computation with the empty family: $!$ of $A \to \top$ has backward part $0^* \to X^*$, and $0^* = 1$ carries only $\varnothing$, whose image is $\varnothing$, which is the backward part of e_A.

For the consequences, both δ_A and the comonoid maps have backward parts built solely from $\cup$ and $\varnothing$, and by Lemma 5.7 each condition to be checked is an equation between two applications of the monoid homomorphism $\alpha^*(u,-)$ to elements of X^* that are equal by associativity, commutativity and unitality of $\cup$. So every such condition holds on the nose rather than as an inequality. A map of free coalgebras $!A \to !B$ is by the co-Kleisli correspondence a morphism $!A \to B$ extended by δ, and its backward part is the monoid homomorphism extending the given map on generators; homomorphisms of commutative monoids commute with $\cup$ and $\varnothing$ by definition, which is precisely the statement that it is a comonoid morphism. □

Theorem 5.9. *For every lineale Ω, $!$ is a linear exponential comonad on $\mathsf{Dial}_\Omega(\mathbf{GrSet})$, which is therefore a model of intuitionistic linear logic, and its co-Kleisli category is cartesian closed.*

Proof. Propositions 5.3, 5.4, 5.5, 5.6 and 5.8 give in turn the functor, the comonad, the Girard isomorphisms, the commutative comonoid structure and its compatibility with the Girard isomorphisms; these are the conditions in the Bierman–Hyland–Schalk characterisation [12, 13]. Section 3 supplies the symmetric monoidal closed structure and the products that the characterisation also requires. □

6 Non-example: vector spaces

The construction of Section 3 was written with one base in mind, the category **Vect** of vector spaces over a field K, with K as truth values. It is symmetric monoidal closed with finite products but not cartesian closed, so it is a base for that construction though not for the one of the thesis, and it is the setting of the best known examples of Chu's construction: $*$-autonomous categories came out of Barr's work on duality for topological vector spaces [24], and Lafont and Streicher observed that finite-dimensional vector spaces sit inside Chu's construction over K, and modelled its modality on that of Dial [9]. The construction of this note is defined over **Vect**, but it behaves differently from Chu's, for a reason that holds over any additive base: in **Vect** finite products are also coproducts.

Lemma 6.1. *In a category with finite biproducts every object carries exactly one monoid structure for the product, namely its addition, with the zero map as unit.*

Proof. A multiplication $m\colon \Omega \oplus \Omega \to \Omega$ is the copairing of $m \circ \iota_1$ and $m \circ \iota_2$, and the unit laws, for the unit $0 \to \Omega$, say that both are the identity. So $m = [\mathrm{id}, \mathrm{id}]$, which is the addition; it is associative and commutative. □

The category **Vect** is additive, so an internal lineale Ω in **Vect** has the addition as multiplication and 0 as unit, the terminal object being the zero space. Its order is equality: an internal partial order on Ω is a subspace R of $\Omega \times \Omega$ containing the diagonal, so $(a, b) \in R$ exactly when $b - a$ lies in the subspace $S = \{c \mid (0, c) \in R\}$, and antisymmetry forces $S \cap (-S) = 0$, that is, $S = 0$. So Ω is a group ordered by equality, with $a \multimap b = b - a$, and every morphism condition of $\mathsf{Dial}_\Omega(\mathbf{Vect})$ is an equation, as in Chu's construction; for $\Omega = K$ this is the group lineale of [7, §8]. An external order does not change this: if $\varphi, \psi\colon W \to \mathbb{R}$ are linear and $\varphi(w) \leq \psi(w)$ for every w, then $\varphi = \psi$, by applying the inequality to $-w$. The witnesses are cocommutative coalgebras and the forward maps coalgebra morphisms.

Proposition 6.2. *Over an additive base $\mathcal{V}$, for any internal lineale Ω, the tensor of $\mathsf{Dial}_\Omega(\mathcal{V})$ is its product, $A \otimes B \cong A \,\&\, B$, and its tensor unit is terminal.*

Proof. Both objects have witnesses $U \otimes V$, and their counterexamples, $X \times Y$ and $X + Y$, coincide because products and coproducts do. By Lemma 6.1 the relation of $A \otimes B$ sends $((u, v), (x, y))$ to $\alpha(u, x)\, \varepsilon(v) + \varepsilon(u)\, \beta(v, y)$, and the relation of $A \,\&\, B$ is the copairing of the same two summands, which is the same map, since a map out of a biproduct is the sum of its restrictions. The units $(I, 1, e)$ and $(I, 0, !)$ coincide because $1 = 0$ and e is the zero map. □

Every object of $\mathsf{Dial}_\Omega(\mathcal{V})$ is then a commutative comonoid, with contraction given by δ_U on witnesses and the addition $X \oplus X \to X$ on counterexamples, and the modality of Section 5 has nothing to add, since the free commutative monoid for the product on X is X itself. The hypothesis of Theorem 3.18 fails in **Vect**, since a linear map between coalgebras need not be a coalgebra map, but it can be replaced. Cocommutative coalgebras over a field form a cartesian closed category [10]; write $\mathrm{Hom}(V, W)$ for its internal hom. Cofree coalgebras also exist [11]. Take $\mathrm{Hom}(U, V) \otimes \sharp[U \otimes Y, X]$ as the witness object of $[A, B]$, with relation defined as in Definition 3.17 using the evaluation of Hom and the counit of $\sharp$. Since $\otimes$ is the product of coalgebras, the forward part of a morphism $A \to [B, C]$ is a coalgebra map $U_A \to \mathrm{Hom}(U_B, U_C)$, that is a coalgebra map $U_A \otimes U_B \to U_C$, together with a coalgebra map $U_A \to \sharp[U_B \otimes Y_C, X_C]$, that is a linear map $U_A \otimes U_B \otimes Y_C \to X_C$; these are the forward data of a morphism $A \otimes B \to C$, the conditions correspond as in the proof of Theorem 3.18, and naturality comes from that of the three adjunctions used. So $\mathsf{Dial}_\Omega(\mathbf{Vect})$ is cartesian closed. Cartesian closure is what one asks of the co-Kleisli category of the modality, where it recovers intuitionistic logic through Girard's translation; here it holds of $\mathsf{Dial}_\Omega(\mathbf{Vect})$ itself, so contraction and weakening hold for every formula, the linear level disappears, and what remains is a model of intuitionistic logic

rather than of linear logic. This is a loss and not a simplification. What linear logic adds to intuitionistic logic is control over contraction and weakening, made visible by the decomposition $A \to B = !A \multimap B$, and in the Dialectica setting that control is where Gödel's interpretation does its work: contraction is the step that forces a choice between counterexamples, which over **Set** the modality records as multisets. In $\mathsf{Dial}_\Omega(\mathbf{Vect})$ contraction is free, two counterexamples being simply added, so the place where the interpretation does its work has disappeared. Over finite-dimensional spaces the cofree coalgebras are not finite-dimensional, so $\mathsf{Dial}_\Omega(\mathbf{FinVect})$ has no internal hom of this form.

The reading of relations as subobjects, set aside in Remark 3.2, avoids Lemma 6.1, since conjunction becomes the intersection of subspaces rather than a map $\Omega \times \Omega \to \Omega$, but it does not give a better category. Take objects (U, X, α) with α a subspace of $U \otimes X$, and morphisms as in Definition 3.5 with containment of pulled-back subspaces in place of the inequality. Intersecting the two pulled-back relations gives a symmetric monoidal structure: since $X \times Y$ is the biproduct $X \oplus Y$ and $\otimes$ preserves coproducts, the counterexample object of $A \otimes B$ splits into two summands, and its relation is the direct sum of the pulled-back relations on them. The unit, with the zero space as counterexamples, is terminal. Products by cases fail, because the backward part of a mediating morphism out of a direct sum adds the contributions of the two summands, so its condition no longer splits: already for $A = B = C = (K, K, 0)$ with identity backward maps, the mediating backward map $(x, y) \mapsto x + y$ pulls $0 \oplus 0$ back to the antidiagonal rather than to 0. Nor have we found an internal hom. Closure would require the relation of $[A, B]$ to be the relative pseudocomplement $P \Rightarrow Q$ of two subspaces of $W \otimes U \otimes Y$, where W is its witness object, and the lattice of subspaces is modular but not distributive: already in K^2, with P a line and $Q = 0$, there is no largest subspace meeting P in Q.

Addition appears in $\mathsf{Chu}_K(\mathbf{Vect})$ too, in the additives, where the pairing on a product is the sum of the two pairings, exactly as for $\mathsf{Dial}_\Omega(\mathbf{Vect})$; the two constructions differ in the tensor. Chu's tensor never combines two truth values, its pairing going through the closed structure of the base, while the Dialectica tensor combines the truth values of its two factors, and over an additive base it can only add them. The distinction is between additive and non-additive bases rather than between finite and infinite dimension, and an additive base never arises in [7], since an additive cartesian closed category is trivial.

Over **Set** the two constructions read vector spaces differently. Sending a vector space V to (V, V^*, ev), with V^* the algebraic dual, embeds **Vect** fully in $\mathsf{Chu}_K(\mathbf{Set})$: the backward map of a Chu morphism does not see the witness, so the condition $\psi(fv) = (g\psi)(v)$ holds uniformly in v, makes every $\psi \circ f$ linear and, the functionals separating points, forces f to be linear. The backward map of a dialectica morphism reads the witness, so in $\mathsf{Dial}_K(\mathbf{Set})$, with K the group lineale of [7, §8], a morphism between such objects needs only $F(v, \psi)(v) = \psi(fv)$ witness by witness, and every function f with $f(0) = 0$ has such backward maps. The Dialectica construction reads a vector space one witness at a time, which is the same freedom that lets its morphisms model Gödel's functionals; linear structure enters it through the base, as in $\mathsf{Dial}_\Omega(\mathbf{Vect})$.

Vector spaces carry several models of linear logic in their own right: Köthe spaces and finiteness spaces [26, 27], convenient vector spaces, due to Blute, Ehrhard and Christine Tasson [28], the Mackey-complete spaces of Marie Kerjean and Christine Tasson [29], and the Fock space models of Blute, Panangaden and Seely [30]. Their categories have finite biproducts, and biproducts are bad for logic, even where, as in quantum computation, they are welcome: they identify the additive conjunction with the additive disjunction, $\& = \oplus$, and they give a zero object, so that every hom-set is inhabited and, read as provability, every formula entails every other. Girard's coherent Banach spaces [31] avoid the first identification, since with contractions as maps products carry the supremum norm and coproducts the ℓ^1 norm, but they keep the zero object, as coherence spaces do. Over a base with biproducts the construction of this note, with relations as maps into an internal lineale, has its tensor equal to its product, but it keeps $\&$ and $\oplus$ apart, $\oplus$ being only a weak coproduct, and it has no zero object, since witnesses are

comonoids and there is no comonoid map from I to 0. The collapse it inherits is of $\otimes$ onto &, not of & onto $\oplus$.

So the construction of Dial over a symmetric monoidal closed category with products works, and $\mathsf{Dial}_\Omega(\mathbf{GrSet})$ is a concrete mathematical example of it. Vector spaces do not seem to be one. Over **Set** the construction does not embed **Vect** fully, since its morphisms read a vector space one witness at a time, and over **Vect** itself it is defined but collapses: with relations as maps into an internal lineale its tensor is its product, and with relations as subspaces its products by cases fail.

7 Conclusions

The Dialectica construction can be carried out over a symmetric monoidal closed base with finite products. What the move off a cartesian base makes visible is the comonoid structure on witnesses and on forward maps, which composition of dialectica morphisms needs and which a cartesian base supplies without being asked. Graded sets give an example over which $\mathsf{Dial}_\Omega(\mathbf{GrSet})$ carries a linear exponential comonad for every lineale Ω, answering over this base the question left open in [3]. Vector spaces, the case we most wanted, do not seem to give the picture we want: over **Vect** the construction is defined and monoidal, but with relations as maps into an internal lineale its tensor is its product, and with relations as subspaces its products by cases fail and we have not found an implication.

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